

Problem 2 (24OIME8)

The answer is 132. It is achieved with $\binom{128}{4}$. We will verify this works in the proof.

We want,

$$\begin{aligned} v_2\left(\binom{n}{k}\right), v_5\left(\binom{n}{k}\right) &\geq 3 \\ \Leftrightarrow \frac{n - s_2(n)}{2 - 1} - \frac{k - s_2(k)}{2 - 1} - \frac{(n - k) - s_2(n - k)}{2 - 1}, \frac{n - s_5(n)}{5 - 1} - \frac{k - s_5(k)}{5 - 1} - \frac{(n - k) - s_5(n - k)}{5 - 1} &\geq 3 \\ \Leftrightarrow \frac{s_2(k) + s_2(n - k) - s_2(n)}{2 - 1} \geq 3; \frac{s_5(k) + s_5(n - k) - s_5(n)}{5 - 1} &\geq 3 \end{aligned}$$

Equivalently, when adding $k, n - k$ in base 2 and base 5 respectively, we carry over at least 3 times. We may verify that $k = 4; n - k = 124$ satisfies this. For this to be possible in base 5, we must have $k + (n - k) = n$ span 4 digits. Therefore, $n \geq 125$.

If $n \geq 131$, then $k > 0$ means $n + k \geq 132$.

If $n = 128, 129, 130$, we may verify that $k \leq 3$ don't lead to the carry-overs in base 5.

If $n = 125, 126, 127$, we may verify that $k \leq 6$ don't lead to the carry-overs in base 2.

Problem 3 (17PTNMB3)

Claim 1: c_n is not "terminating" -- there is no threshold k such that $c_i \equiv 0$ for all $i \geq k$ or $c_i \equiv 1$ for all $i \geq k$.

For a "terminating" sequence c_n , define its "terminus" as the minimum k such that $c_i \equiv 0$ for all $i \geq k$ or $c_i \equiv 1$ for all $i \geq k$. If the "terminus" is zero, we may check that neither $c_n \equiv 1$ nor $c_n \equiv 0$ work.

In the case where $c_i \equiv 0$ after the terminus, we take v_3 of the sum $f\left(\frac{2}{3}\right)$ of a finite number of fractions to get $v_3\left(f\left(\frac{2}{3}\right)\right) \leq 0$. In the case where $c_i \equiv 1$ after the terminus k , we may take the geometric series $c_k\left(\frac{2}{3}\right)^k + c_{k+1}\left(\frac{2}{3}\right)^{k+1} + \dots = 3\left(\frac{2}{3}\right)^k$. Since k is the terminus, $c_{k-1} = 0$ meaning the total sum would have $v_3\left(f\left(\frac{2}{3}\right)\right) = -(k - 1) \leq 0$. In all cases, $v_3\left(f\left(\frac{2}{3}\right)\right) \neq v_3\left(\frac{3}{2}\right) = 1$. $\square$

FTSOC, let $f\left(\frac{1}{2}\right) = \frac{p}{q}$ be rational.

Claim 2: If $f\left(\frac{1}{2}\right) = \frac{p}{q}$ for some fixed $\frac{p}{q}$, c_n must be unique.

Sketch: Note that c_n is terminating. Use FTSOC then bounding fails.

Claim 3: If $f\left(\frac{1}{2}\right)$ is rational, c_n must be repeating after a certain threshold.

WLOG, let $\gcd(p, q) = 1$. We first prove this is true for odd q and $p < q$. Note that $2^{\varphi(q)} - 1 \equiv 0 \pmod q$. Therefore, we may write $\frac{p}{q} = \frac{pk}{2^{\varphi(q)} - 1}$ for some integer k . We let $c_0 = 1$ and $c_1, \dots, c_{\varphi(q)}$ be the expression of pk in binary. We then repeat $c_n = c_{n-\varphi(q)}$ to finish. Since c_n is unique, we are done. If $1 \leq \frac{p}{q} < 2$. We may do the same thing except with $c_0 = 1$.

If q is even, let k be the integer such that $v_2\left(f\left(\frac{1}{2}\right) \cdot 2^k\right) = 0$. We may then see that $f\left(\frac{1}{2}\right) \cdot 2^k - \left\lfloor f\left(\frac{1}{2}\right) \cdot 2^k \right\rfloor$ is a fraction with an odd denominator under 1 which will repeat. Therefore, since $\left\lfloor f\left(\frac{1}{2}\right) \cdot 2^k \right\rfloor$ is just a whole number, the binary representation of $f\left(\frac{1}{2}\right) \cdot 2^k$. If we bit shift, we see that the binary representation of $f\left(\frac{1}{2}\right)$ is repeating. Therefore, the c_n that matches to it is repeating. $\square$

If we let the eventual period of c_n be p and have it start repeating after index s . We see that, for some integers m, n ,

$$f\left(\frac{2}{3}\right) = \sum_{i=0}^s c_i \left(\frac{2}{3}\right)^i + \frac{\sum_{i=s+1}^{s+p} c_i \left(\frac{2}{3}\right)^i}{1 - \left(\frac{2}{3}\right)^p}$$

Taking v_2 , we find $\sum_{i=0}^s c_i \left(\frac{2}{3}\right)^i$ is either zero or $v_2\left(\sum_{i=0}^s c_i \left(\frac{2}{3}\right)^i\right) \geq 0$. Same thing goes for $\frac{\sum_{i=s+1}^{s+p} c_i \left(\frac{2}{3}\right)^i}{1 - \left(\frac{2}{3}\right)^p}$.

Therefore, we cannot get $v_2\left(\frac{2}{3}\right) = -1$ in the end.

6 clubs; 2.5 hours

Problem 4 (19TSTST7)

[All the other solutions are using modular arithmetic making this one feel too simple. I'm worried but cannot find any errors]

Note that since the range of f is finite, by the well-ordering principle, we may find a pair of integers m, n such that $f(-m) = n$ where n is the maximum of f . We claim that,

$$f(x) \equiv \gcd(m + x, n)$$

Note that, if $f(x)$ works as a solution, $g(x) = f(x + k)$ works as a solution for any integer k . Therefore, WLOG, let $m = 0$.

Claim 1: For all primes p , $v_p(f(x)) \geq \min(v_p(x), v_p(n))$ for all x .

Proof: Plugging in $(0, x)$ into the equation, $\gcd(n, f(x)) = \gcd(n, x)$.

Therefore, $\min(v_p(f(x)), v_p(n)) = \min(v_p(n), v_p(x))$ which means $v_p(f(x)) < \min(v_p(x), v_p(n))$ is impossible.

Claim 2: For all primes p , $v_p(f(x)) \leq \min(v_p(x), v_p(n))$ for all x .

Proof: FTSOC, let there be a prime p and integer x such that $v_p(f(x)) > \min(v_p(x), v_p(n))$. Using Claim 1, we still get $v_{p'}(f(x)) \geq \min(v_p(x), v_p(n))$ for all other primes p' . Therefore, we get $f(x) > n$ which contradicts our assumption that n is the maximum in our range.

In conclusion, $v_p(f(x)) = \min(v_p(x), v_p(n))$ as desired and we are done.

Problem 5 (16AMO2)

Taking v_p for an arbitrary prime,

$$v_p \left((k^2)! \cdot \prod_{j=0}^{k-1} \frac{j!}{(j+k)!} \right) \geq 0$$

$$\Leftrightarrow \sum_{i>0} \left\lfloor \frac{k^2}{p^i} \right\rfloor + \sum_{j=0}^{k-1} \sum_{i>0} \left\lfloor \frac{j}{p^i} \right\rfloor - \sum_{i>0} \left\lfloor \frac{j+k}{p^i} \right\rfloor \geq 0$$

We show this is true for each i ,

$$\Leftrightarrow \left\lfloor \frac{k^2}{p^i} \right\rfloor \geq \sum_{j=0}^{k-1} \left\lfloor \frac{j+k}{p^i} \right\rfloor - \left\lfloor \frac{j}{p^i} \right\rfloor$$

Let $k = p^i x + y$.

$$\Leftrightarrow \left\lfloor \frac{p^{2i}x^2 + 2p^i xy + y^2}{p^i} \right\rfloor \geq \sum_{j=0}^{k-1} \left\lfloor \frac{j + p^i x + y}{p^i} \right\rfloor - \left\lfloor \frac{j}{p^i} \right\rfloor$$

$$\Leftrightarrow p^i x^2 + 2xy + \left\lfloor \frac{y^2}{p^i} \right\rfloor \geq \sum_{j=0}^{p^i x + y - 1} x + \left\lfloor \frac{j+y}{p^i} \right\rfloor - \left\lfloor \frac{j}{p^i} \right\rfloor$$

$$\Leftrightarrow xy + \left\lfloor \frac{y^2}{p^i} \right\rfloor \geq \sum_{j=0}^{p^i x + y - 1} \left\lfloor \frac{j+y}{p^i} \right\rfloor - \left\lfloor \frac{j}{p^i} \right\rfloor$$

$$\Leftrightarrow xy + \left\lfloor \frac{y^2}{p^i} \right\rfloor \geq \sum_{j=0}^{p^i x - 1} \left\lfloor \frac{j+y}{p^i} \right\rfloor - \left\lfloor \frac{j}{p^i} \right\rfloor + \sum_{j=p^i x}^{p^i x + y - 1} \left\lfloor \frac{j+y}{p^i} \right\rfloor - \left\lfloor \frac{j}{p^i} \right\rfloor$$

$$\Leftrightarrow \left\lfloor \frac{y^2}{p^i} \right\rfloor \geq \sum_{j=p^i x}^{p^i x + y - 1} \left\lfloor \frac{j+y}{p^i} \right\rfloor - \left\lfloor \frac{j}{p^i} \right\rfloor$$

$$\Leftrightarrow \left\lfloor \frac{y^2}{p^i} \right\rfloor \geq \max(0, 2y - p^i)$$

Note that the LHS is clearly greater than or equal to zero,

$$\Leftrightarrow \left\lfloor \frac{y^2}{p^i} \right\rfloor \geq 2y - p^i$$

We prove this using induction.

Base case: $y = p^i - 1$

The LHS would be $\left\lfloor \frac{p^{2i} - 2p^{i+1} + 1}{p^i} \right\rfloor = p^i - 2$ and RHS is $2(p^i - 1) - p^i = p^i - 2$ also.

Inductive step: $y \rightarrow y - 1$

Note that, clearly, since $y < p^i$,

$$\left\lfloor \frac{y^2}{p^i} \right\rfloor - \left\lfloor \frac{(y-1)^2}{p^i} \right\rfloor = \left\lfloor \frac{y^2}{p^i} \right\rfloor - \left\lfloor \frac{y^2 - 2y + 1}{p^i} \right\rfloor \leq 2$$

Conclusion: By induction, we are done.

14 clubs; 5 hours

Problem 7 (18CHNTST12)

We claim the answer is all numbers n such that, for some prime p , $v_p(n) \geq 1009$ except $n = 2^{1009}$.

To prove attainability, we write $n = p^{1009}(m)$ for some positive integer m . Since $pm > 2$, there is some $k \bmod pm$ that is not a quadratic residue and $\gcd(k, pm) = 1$. We may choose the sequence,

$$(kp^{1008}), (kp^{1008}) + (p^{1009}m), (kp^{1008}) + 2(p^{1009}m), (kp^{1008}) + 3(p^{1009}m), \dots$$

Note that the numbers in this sequence are of the form,

$$kp^{1008} + x(p^{1009}m) = p^{1008}(k + xpm)$$

Since p does not divide k , p does not divide $k + xpm$. Therefore,

$$d(p^{1008}(k + xpm)) = d(p^{1008}) \cdot d(k + xpm) = 1009 \cdot d(k + xpm)$$

Since $k + xpm \equiv k \bmod pm$ is not a quadratic residue, $k + xpm$ is not a perfect square meaning we have $2 \mid d(k + xpm)$.

We will prove that 2^{1009} fails at the end.

To prove that nothing else works, FTSOC let there be a start x of the sequence with common difference $n = p_1^{e_1} p_2^{e_2} \dots p_o^{e_o}$ such that $e_1, e_2, \dots$ are all less than 1009. Let $g := \gcd(n, x) = p_1^{e'_1} p_2^{e'_2} \dots p_o^{e'_o}$. WLOG,

let $p_1, \dots, p_i$ be the primes in the factorization of g with multiplicity 1008. Note that for each of these primes p_j , $v_{p_j}(n) = v_{p_j}(x) = 1008$ for $v_{p_j}(g) = 1008$ to happen.

Claim: For some integer k , all other integer $k' \equiv k \pmod{p_j^2}$ satisfy $v_{p_j}(x + k'n) = 1009$.

Proof: Let $N = \frac{n}{p_j^{1008}}; X = \frac{x}{p_j^{1008}}$. We prove that $v_{p_j}(X + k'N) = 1$ exactly for some $k' \pmod{p_j^2}$. Firstly, out of $x + n, x + 2n, \dots, x + p_j n$, there is some selection such that $p_j \mid x + mn$ since both X, N are not divisible by p . This is true for all $m' \equiv m \pmod{p_j}$. Out of,

$$x + (m + p_j)n, x + (m + 2p_j)n, \dots, x + (m + p_j^2)n$$

There exists exactly one $x + (m + m_2 p_j)n$ such that $p_j^2 \mid x + (m + m_2 p_j)n$. All the others satisfy $v_{p_j} = 1$ exactly and are options for k . $\square$

If we do this for $p_1, \dots, p_i$, by CRT, we will find some integer k where all $k' \equiv k \pmod{p_1^2 \dots p_i^2}$ satisfy $v_{p_1}(x + k'n) = \dots = v_{p_i}(x + k'n) = 1009$. [Citation needed from claim to now?] We write, for some y ,

$$x + k'n = x + (k + p_1^2 \dots p_i^2 y)n = (x + kn) + (p_1^2 \dots p_i^2 y)n$$

We now let $g_1 := \gcd(x + kn, p_1^2 p_2^2 \dots p_i^2 n)$. Note that $v_{p_{i+1}}(g_1), v_{p_{i+2}}(g_1), \dots, v_{p_o}(g_1) < 1008$ since $v_{p_{i+1}}(p_1^2 p_2^2 \dots p_i^2 n) < 1008$ and so on.

By Dirichlet, the sequence $\frac{(x+kn)}{g_1}, \frac{(x+kn)}{g_1} + \frac{(p_1^2 \dots p_i^2 n)}{g_1}, \frac{(x+kn)}{g_1} + 2 \cdot \frac{(p_1^2 \dots p_i^2 n)}{g_1}, \dots$ has infinitely many primes. Let $\frac{(x+kn)}{g_1} + m_2 \cdot \frac{(x+kn)}{g_1} = p'$ be a prime relatively prime to x, n for some m_2 .

Therefore, $(x + kn) + (p_1^2 \dots p_i^2 n)m_2 = p'g$ is in our arithmetic sequence. However, $d(p'g) = 2d(g)$. Since $v_{p_1}(g) = v_{p_2}(g) = \dots = v_{p_i}(g) = 1009$ and $v_{p_{i+1}}(g) = \dots = v_{p_o}(g) < 1008$, but 1009 is prime, it is impossible for $1009 \mid d(p'g)$ meaning we have a contradiction.

Now to prove 2^{1009} fails, let x be the start once again. Once again, if $v_2(x) \neq 1008$, we finish using Dirichlet. Let $x = 2^{1008}X$ for some odd number X . Our sequence would be, for some integer k ,

$$2^{1008}(X + 2k)$$

Since there are infinitely many squares, we may find some k such that $X + 2k$ is an odd square.

$$d(2^{1008}) \cdot d(X + 2k) = 1009 \cdot d(X + 2k)$$

Since $X + 2k$ is a square, $d(X + 2k)$ is odd meaning $d(2^{1008}(X + 2k))$ is odd and cannot be 2018.

Problem 9 (14TSTST6)

Note that $p \mid ca^n - db^n \Leftrightarrow \frac{a}{c} \equiv \left(\frac{a}{b}\right)^n \pmod{p}$. This must occur for some n according to the problem statement. Thus, if we let $t = \text{ord}_p(n)$ and s be the smallest positive integer n such that this is true, this occurs if and only if $n = s + kt$ for some k . We find that,

$$v_p(ca^{s+kt} - db^{s+kt}) = v_p\left(\left(\frac{a^t}{b^t}\right)^k - \frac{db^s}{ca^s}\right)$$

We may substitute in $x = \frac{a^t}{b^t}; y = \frac{db^s}{ca^s}$. Note that $x = \frac{a^t}{b^t} \equiv 1 \pmod{p}$ meaning $y = \frac{db^s}{ca^s} \equiv 1 \pmod{p}$ also for the difference to be divisible by p . We show that, if for some pair of integers m, n , if,

$$d = v_p(x^m - y) < v_p(x^n - y) = e$$

We may find some α such that $v_p(x^\alpha - y) > e$ which implies that v_p is unbounded by induction. This would finish the problem. Note that,

$$\begin{aligned} v_p((x^m - y) - (x^n - y)) &= d \\ \Leftrightarrow v_p(x^{|m-n|} - 1) &= d \end{aligned}$$

By LTE,

$$\Rightarrow v_p(x^{p^{e-d}|m-n|} - 1) = e$$

We let $k := p^{e-d}|m-n|$. Note that for some quotients u, v such that $v_p(u) = v_p(v) = 0$,

$$\begin{aligned} x^k &= p^e u + 1; \\ x^m &= p^e v + y \end{aligned}$$

For any integer $1 \leq r \leq p-1$,

$$\begin{aligned} x^{kr+m} - y &= (p^e u + 1)^r \cdot (p^e v + y) - y \\ &= p^e(v + yur) + p^{2e}(\dots) \end{aligned}$$

There exists a choice for r such that $v + yur \equiv 0 \pmod{p}$. The resulting sum will have v_p at least $e+1$ as desired.

24 clubs; 9 hours

Problem 10 (17ESLN3)

$$\text{Lemma: } 1 + \left(\frac{1 + \left(1 + \frac{1}{2}\right) + \dots + \left(1 + \frac{1}{2} + \dots + \frac{1}{k-1}\right)}{k} \right) = 1 + \frac{1}{2} + \dots + \frac{1}{k}$$

Proof: We prove this by induction.

Base case: Trivially, for $k = 1$, $1 + 0 = 1$.

Inductive step: We have,

$$1 + \left(\frac{1 + \left(1 + \frac{1}{2}\right) + \dots + \left(1 + \frac{1}{2} + \dots + \frac{1}{k-1}\right)}{k} \right) = 1 + \frac{1}{2} + \dots + \frac{1}{k}$$

$$\begin{aligned}
&\Leftrightarrow 1 + \left(\frac{1 + \left(1 + \frac{1}{2}\right) + \dots + \left(1 + \frac{1}{2} + \dots + \frac{1}{k}\right)}{k} \right) = \left(1 + \frac{1}{2} + \dots + \frac{1}{k}\right) + \frac{\left(1 + \frac{1}{2} + \dots + \frac{1}{k}\right)}{k} \\
&\Leftrightarrow \frac{k}{k+1} + \left(\frac{1 + \left(1 + \frac{1}{2}\right) + \dots + \left(1 + \frac{1}{2} + \dots + \frac{1}{k}\right)}{k+1} \right) = \left(1 + \frac{1}{2} + \dots + \frac{1}{k}\right) \\
&\Leftrightarrow 1 + \left(\frac{1 + \left(1 + \frac{1}{2}\right) + \dots + \left(1 + \frac{1}{2} + \dots + \frac{1}{k}\right)}{k+1} \right) = \left(1 + \frac{1}{2} + \dots + \frac{1}{k+1}\right)
\end{aligned}$$

Conclusion: By mathematical induction we are done. $\square$

We claim that no C work. Let $C = p_1^{e_1} p_2^{e_2} \dots p_n^{e_n}$.

Claim: For each i, m , we have $v_{p_i}(a_m) \leq v_{p_i}(a_1) + e_i \left(\sum_{j=1}^{m-1} \frac{1}{j} \right)$.

Proof: We prove this by strong induction on m .

Base case: Clearly, for $m = 1$, $v_{p_i}(a_1) \leq v_{p_i}(a_1) + 0$.

Inductive step: We have,

$$a_{k+1}^k \mid C^k a_1 a_2 \dots a_k$$

Taking v_{p_i} ,

$$\Rightarrow k v_{p_i}(a_{k+1}) \leq k e_i + v_{p_i}(a_1) + \dots + v_{p_i}(a_k)$$

Assuming the RHS is maximized,

$$\begin{aligned}
&\Rightarrow k v_{p_i}(a_{k+1}) \leq k e_i + (v_{p_i}(a_1) + 0) + (v_{p_i}(a_1) + e_i(1)) + \dots \\
&\quad + \left(v_{p_i}(a_1) + e_i \left(1 + \frac{1}{2} + \dots + \frac{1}{k-1} \right) \right) \\
&\Leftrightarrow v_{p_i}(a_{k+1}) \leq v_{p_i}(a_1) + e_i \left(1 + \frac{1 + \left(1 + \frac{1}{2}\right) + \dots + \left(1 + \frac{1}{2} + \dots + \frac{1}{k-1}\right)}{k} \right) \\
&\Leftrightarrow v_{p_i}(a_{k+1}) \leq v_{p_i}(a_1) + e_i \left(1 + \frac{1}{2} + \dots + \frac{1}{k} \right) \square
\end{aligned}$$

Conclusion: By induction we are done. $\square$

Note that $v_{p_i}(a_1) + e_i \left(\sum_{j=1}^{m-1} \frac{1}{j} \right)$ is an increasing function in m . Therefore, for a given l , it is true that $v_{p_i}(a_m) \leq v_{p_i}(a_1) + e_i \left(\sum_{j=1}^{l-1} \frac{1}{j} \right)$ for any $m \leq l$; i . Thus, since $a_1, \dots, a_l$ are distinct, we have,

$$l \leq \prod_{1 \leq i \leq n} \left(1 + v_{p_i}(a_1) + e_i \left(\sum_{j=1}^{l-1} \frac{1}{j} \right) \right)$$

Where the RHS represents the distinct options for each $a_1, \dots, a_l$ satisfying the v_{p_i} requirements.

Using harmonic sums, $\sum_{j=1}^{l-1} \frac{1}{j} \leq 1 + \ln(l-1)$.

$$\Rightarrow l \leq \prod_{1 \leq i \leq n} \left(1 + v_{p_i}(a_1) + e_i(1 + \ln(l)) \right)$$

We now apply big-O notation [citation needed?],

$$\Rightarrow O(l) \leq O \left(\prod_{1 \leq i \leq n} \left(1 + v_{p_i}(a_1) + e_i(1 + \ln(l)) \right) \right)$$

Noting that $n, v_{p_i}(a_1), 1, e_i$ are constants, if we expand out the terms of the RHS, we'd get a degree- n polynomial in $\ln(l)$ where all the coefficients are constants. Therefore,

$$\Leftrightarrow O(l) \leq O(\ln(l)^n)$$

However, as l approaches infinity, we'd eventually have $l > \ln(l)^n$ [citation needed?] which is a contradiction as desired.

Problem 13 (H3018301)

We may quickly check that 6, 69, 696, 6969 are clearly not squares.

For the sake of contradiction, let there exists some $m > 264$ such that, $m^2 = 69696 \dots$

Case 1: m^2 ends with 6.

We may then write,

$$(m - 264)(m + 264) = m^2 - 264^2 = 69 \dots 69000000$$

Note that v_2 of the RHS is clearly 5. However, since $v_2(264) = 3$, either v_2 of the LHS is even if $v_2(m) < 3$ or greater than or equal to 6. Either case, it cannot be 5. (!)

Case 2: m^2 ends with 9.

We let, $m^2 = 6969 \dots 69 = 69 \cdot (10101 \dots 01) = 69 \cdot \frac{100^k - 1}{100 - 1}$ for some positive k .

$$\Leftrightarrow m^2 = \frac{23}{33} \cdot (10^{2k} - 1) = \frac{23}{33} \cdot (10^k - 1)(10^k + 1)$$

$$\Leftrightarrow m^2 = \frac{23}{33} \cdot (10^{2k} - 1) = \frac{23}{33} \cdot (10^k - 1)(10^k + 1)$$

This is possible if and only if, for all primes p ,

$$\begin{aligned} v_p(m^2) &= v_p\left(\frac{23}{33}\right) + v_p(10^k - 1) + v_p(10^k + 1) \\ \Rightarrow 2 \mid v_p\left(\frac{23}{33}\right) + v_p(10^k - 1) + v_p(10^k + 1) \end{aligned}$$

For $p = 3$, note that $v_3(10^k + 1) = 0$ always since $10^k + 1 \equiv 1^k + 1 \equiv 2 \pmod{3}$. Since $v_p\left(\frac{23}{33}\right) = -1$, we must have $v_3(10^k - 1)$ to be odd and the other is zero.

For $p = 11$, we see that $11 \mid 10^k - 1$ for even k and $11 \mid 10^k + 1$ for odd k . Thus, for even k , we must have $v_{11}(10^k - 1)$ to be odd. For odd k , we must have $v_{11}(10^k + 1)$ to be odd and the other is zero.

For $p = 23$, note that $\text{ord}_{23}(10) \mid 22$. We clearly have $10^1, 10^2 \not\equiv 1 \pmod{23}$. Additionally, we'd get $10^{11} \equiv 2^{11} \cdot 5^{11} \equiv (2 \cdot 9^2) \cdot (5 \cdot 2^5) \equiv -1 \pmod{23}$ so 11 doesn't work. Therefore, $\text{ord}_{23}(10) = 22$. If $k \equiv 11 \pmod{22}$ is odd, then $v_{11}(10^k + 1)$ must be odd. If $k \equiv 0 \pmod{22}$ is even, then $v_{11}(10^k - 1)$ must be odd and the other is zero.

For all other primes p , one of $v_p(10^k - 1)$ and $v_p(10^k + 1)$ is even and the other is zero since $\gcd(10^k - 1, 10^k + 1) = 2$.

In conclusion, if k is even, we may write $10^k - 1 = 3(11)(23)y^2$; $10^k + 1 = x^2$ for some x, y integers.

If k is odd, we may write $10^k - 1 = 3y^2$; $10^k + 1 = 11(23)x^2$ for some x, y integers.

In either case, we'd need $10^k + 1 \equiv x^2 \pmod{3}$. However, $10^k + 1 \equiv 2 \pmod{3}$. (!)

32 clubs; 13 hours

Problem 15 (18IMO5)

For all $n \geq N$, we have both,

$$\begin{aligned} \frac{a_1}{a_2} + \frac{a_2}{a_3} + \cdots + \frac{a_{n-1}}{a_n} + \frac{a_n}{a_1} &\in \mathbb{Z} \\ \frac{a_1}{a_2} + \frac{a_2}{a_3} + \cdots + \frac{a_n}{a_{n+1}} + \frac{a_{n+1}}{a_1} &\in \mathbb{Z} \end{aligned}$$

Taking the difference,

$$\frac{a_n}{a_{n+1}} + \frac{a_{n+1} - a_n}{a_1} \in \mathbb{Z}$$

For any prime p , we claim that $v_p(a_n)$ eventually becomes constant. Consider if $v_p(a_n) \neq v_p(a_{n+1})$.

Case 1: $v_p(a_n) < v_p(a_{n+1})$.

We have $v_p\left(\frac{a_n}{a_{n+1}}\right) = v_p(a_n) - v_p(a_{n+1}) = 0$ meaning we must also have,

$$v_p\left(\frac{a_{n+1} - a_n}{a_1}\right) = v_p(a_n) - v_p(a_{n+1})$$

However, $v_p(a_{n+1} - a_n) = v_p(a_n)$ so,

$$\begin{aligned} \Leftrightarrow v_p(a_n) - v_p(a_1) &= v_p(a_n) - v_p(a_{n+1}) \\ \Leftrightarrow v_p(a_{n+1}) &= v_p(a_1) \end{aligned}$$

Case 2: $v_p(a_n) > v_p(a_{n+1})$

We have $v_p\left(\frac{a_n}{a_{n+1}}\right) > 0$ in this case meaning,

$$v_p\left(\frac{a_{n+1} - a_n}{a_1}\right) \geq 0 \Leftrightarrow v_p(a_{n+1}) \geq v_p(a_1)$$

We let, for $x_1 < x_2 < \dots$ which are either integers or positive infinity,

$$\begin{aligned} v_p(a_N) &= \dots = v_p(a_{N+x_1}) = b_1; \\ v_p(a_{N+x_1+1}) &= \dots = v_p(a_{N+x_2}) = b_2; \\ &\dots \end{aligned}$$

Such that $b_1 \neq b_2 \neq b_3 \neq \dots$. We prove that b_i has a finite number of terms.

From the casework we did before, if $b_i < b_{i+1}$, we have $b_{i+1} = v_p(a_1)$. If $b_i > b_{i+1}$, we have $b_i > b_{i+1} > v_p(a_1)$. In either case, $|b_i - v_p(a_1)|$ will strictly decrease until it hits zero at which point the sequence ends.

After b_i , all terms $v_p(a_{N+x_{i-1}+1}), v_p(a_{N+x_{i-1}+2}), \dots$ must stay equal to $v_p(a_1)$ meaning it becomes constant.

If we apply this to each prime, we see that after a certain threshold, $v_p(a_k) = v_p(a_{k+1}) = \dots$ for all primes p in which case (a_n) must have gone constant.

Problem 16 (09AMO6)

Note that we may increment each term of s_i by $-s_0$ and increment each term of t_i by $-t_0$ and not affect our result. This way, $s_1 = t_1 = 0$. Therefore,

$$(s_i - 0)(t_i - 0) = s_i t_i \in \mathbb{Z}$$

We further get,

$$\begin{aligned} (s_i - s_j)(t_i - t_j) &= s_i t_i - s_j t_i - s_i t_j + s_j t_j \in \mathbb{Z} \\ \Leftrightarrow s_i t_j + s_j t_i &\in \mathbb{Z} \end{aligned}$$

Since s is not constant, we may make a swap such that $s_2 \neq 0$. We now multiply each term in s_i by $\frac{1}{s_2}$ and each term in t_i by s_2 . This also does not affect us since $(s_i - s_j)(t_i - t_j)$ once again remains constant and r still works if we multiply it by s_2 also. This way, $s_2 = 1$.

We let $r_2 := x$. Note that x is an integer since $s_2 r_2$ is an integer. For all $i > 2$, let,

$$s_i = \frac{a_i}{b_i} \text{ for } \gcd(a_i, b_i) = 1$$

Therefore, for some integer y_i ,

$$t_i = y_i \cdot \frac{b_i}{a_i}$$

Note that,

$$s_2 t_i + s_i t_2 \in \mathbb{Z} \Leftrightarrow y_i \cdot \frac{b_i}{a_i} + x \cdot \frac{a_i}{b_i} \in \mathbb{Z} \Leftrightarrow a_i b_i \mid y_i b_i^2 + x a_i^2$$

Since $\gcd(a_i, b_i) = 1$,

$$\Leftrightarrow a_i \mid y_i; b_i \mid x$$

Notice that $\text{lcm}(b_3, b_4, \dots) \mid x$. We multiply each term of s_i by $\text{lcm}(b_3, b_4, \dots)$ and divide each term of t_i by this lcm.

Claim: $t_i \in \mathbb{Z}$ for all i .

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Problem 19 (24JMO3)

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Claim 1: For any positive integer m , either $a_{m(p-1)-1}$ or $a_{m(p-1)}$ are divisible by p .

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Proof: If $p \nmid a_{m(p-1)-1}$, $a_{m(p-1)} \equiv a_{m(p-1)-1}^{m(p-1)} - 1 \equiv 0 \pmod{p}$. $\square$

We split into two cases.

Case 1: $a_n \equiv 0 \pmod{p}$ for some even n .

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Note that $a_{n+1} \equiv a_n^{n+1} - 1 \equiv -1 \pmod{p}$ meaning $a_{n+2} \equiv a_{n+1}^{n+2} - 1 \equiv 0 \pmod{p}$. Therefore, the terms modulo p will go back and forth between -1 and 0 . Let $K > k$ be an integer such that $p^K > n$,

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$$a_{2p^K} = a_{2p^{K-1}}^{2p^K} - 1 = \left(a_{2p^{K-1}}^2\right)^{p^K} - 1^{p^K}$$

.

By LTE, since $a_{2p^{K-1}}^2 \equiv 1 \pmod{p}$,

$$v_p(a_{2p^K}) > v_p(2p^K) = K > k$$

Case 2: $a_n \equiv 0 \pmod{p}$ occurs for odd n only.

By claim 1, since $p - 1$ is odd, we must have $p \mid a_{m(p-1)-1}$ for all m . Thus,

$$a_{m(p-1)-1} \equiv a_{m(p-1)-2}^{m(p-1)-1} - 1 \equiv 0 \pmod{p}$$

$$\Leftrightarrow a_{m(p-1)-2}^{-1} \equiv 1 \pmod{p}$$

$$\Leftrightarrow a_{m(p-1)-2} \equiv 1 \pmod{p}$$

Note that $(p-1)p^k \equiv -1 \pmod{p-1}$ meaning $\exists m$ such that $(p-1)p^k = m(p-1) - 1$. For this m ,

$$v_p(a_{m(p-1)-1}) = v_p(a_{m(p-1)-2}^{(p-1)p^k} - 1^{(p-1)p^k})$$

Once again, by LTE, we are done.

51 clubs; 16.5 hours